MODULE 05 SESSION 18

The model: a star, relationships, a date table

One model that answers revenue by customer, product or month.

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Wed 4 and Sun 8 Nov 2026

DATA SCIENCE & ANALYTICS

BY THE END OF TODAY, YOU CAN

Class objectives: build and read a star model

01 Explain facts, dimensions and the star

02 Name the grain of each table

03 Build and mark a date table

04 Relate tables; read cardinality and direction

05 Use the delivered date and tidy the model

PART 1

What a Power BI model is, and why its tables form a star

Before we build Nakheel's model: what a model is, and the shape that makes it easy to ask questions of.

INSTRUCTOR DO

Step 1.1

What a model is, and the two jobs of sales data

A model is tables and the relationships between them. Then a café: its sales are recorded at the till and used to answer questions.

A model is the tables plus the relationships between them

SO FAR · D16 AND D17 · EACH VISUAL READ ONE TABLE





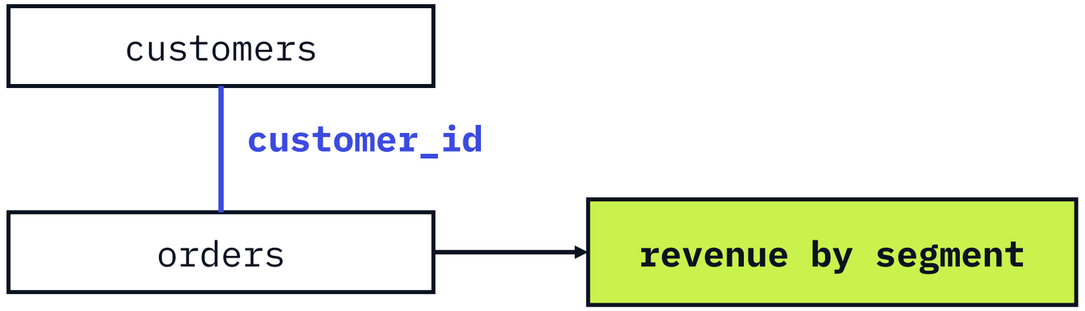
a column from another table meant a merge first

BIGQUERY · D06’S LEFT JOIN, BY SEGMENT

```
SELECT c.segment, SUM(o.order_total)
FROM nakheel.orders AS o
LEFT JOIN nakheel.customers AS c
  ON o.customer_id = c.customer_id
GROUP BY c.segment
```

you write the join in every query

POWER BI · THE SAME JOIN, SAVED ONCE



one visual, columns from both tables

A relationship is a join saved in the model; Power BI calls the model a semantic model and runs the join when a visual needs it.

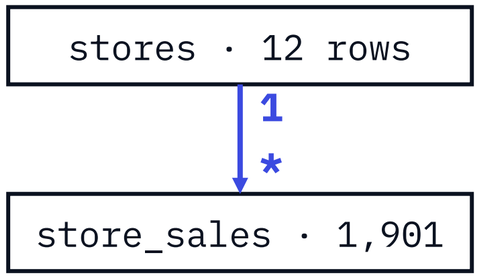
What Merge and a relationship each do with two tables

D17 · MERGE · store_sales_by_store

store_code	order_total	governorate	area
NK-AMM-01	47.00	Amman	Abdali
NK-KRK-01	22.50	Karak	Karak

indigo: the store's columns, copied onto all 1,901 sales

TODAY · A RELATIONSHIP ON store_code



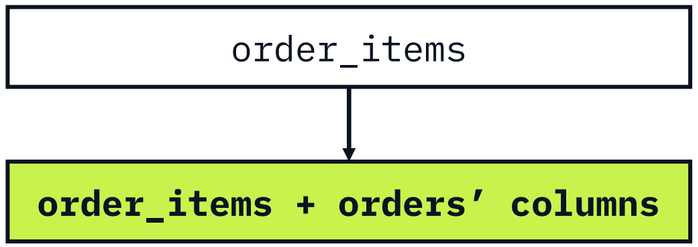
a relationship: a line on the shared key. Each column lives once; the line joins them when a visual asks

What does each give a report?

Revenue by governorate	both: the same nine bars (Amman 83,914.50 … (Blank) 858.00), total 205,338.00	✓
The store list gains a column	merge: redo the expand · relationship: it is there at once	✓
Where the join happens	merge: Power Query, at every refresh · relationship: the model, when a visual asks	✓

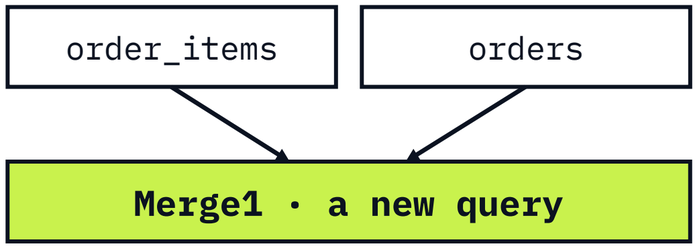
What each way of combining tables leaves in the model

MERGE QUERIES



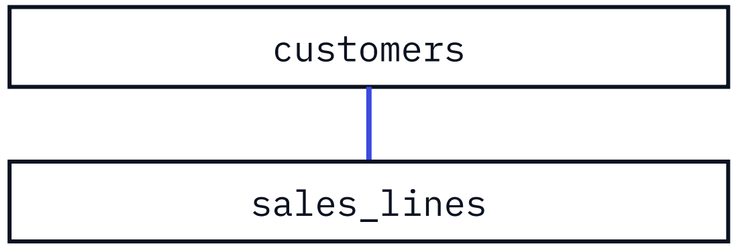
the same query gets wider

MERGE QUERIES AS NEW



a new table, loaded unless you turn load off

A RELATIONSHIP



no new table; nothing is copied

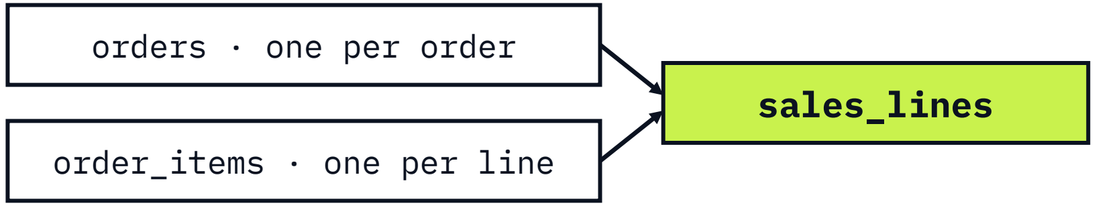
	Merge	Merge as New	Relationship
A new table in the model	no	yes, unless load is off	**no**
Columns copied onto rows	yes	yes	**no**
When the join runs	every refresh	every refresh	**when a visual asks**

Merge changes the tables. A relationship only connects them.

A RULE OF THUMB · WITH THE TABLES OF SUNDAY'S MODEL

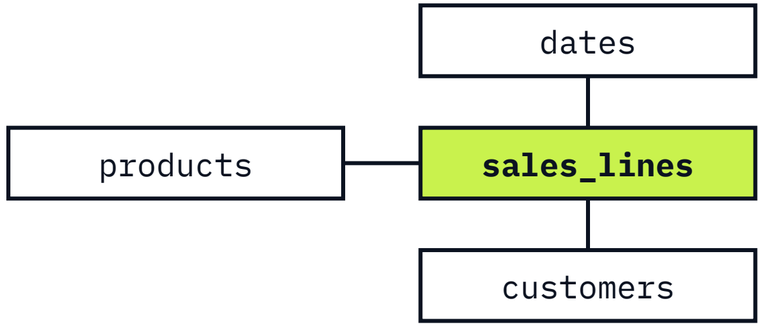
When to use Merge and when to use a relationship

MERGE · BUILD ONE TABLE AT ONE GRAIN



each line has one order, so the result keeps one row per line

RELATIONSHIP · CONNECT A DIMENSION TO A FACT



one row in the dimension, many in the fact

Warning sign: a Merge that copies a description onto many rows, like D17's governorate on all 1,901 sales. Use a relationship instead.

Merge to build a table. Use a relationship to connect tables.

MADE-UP CAFÉ · THREE BRANCHES · 9,180 RECEIPT LINES IN OCTOBER 2026

Recording sales and answering questions use the same data

JOB 1 · RECORD A SALE AT THE TILL

```
Receipt 4127 · Irbid
18 Oct, 08:42

2 × Latte      5.00
1 × Espresso   1.50
1 × Mint tea   1.25
Total          7.75
```

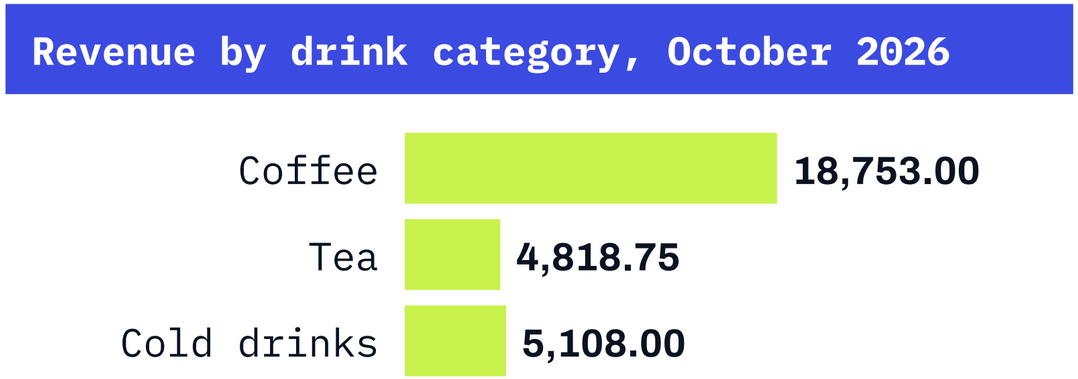
→

receipt_lines

drink	qty	amount
Latte	2	5.00
Espresso	1	1.50
Mint tea	1	1.25

writes 3 rows, one receipt at a time, all day

JOB 2 · ANSWER THE MANAGER'S QUESTION



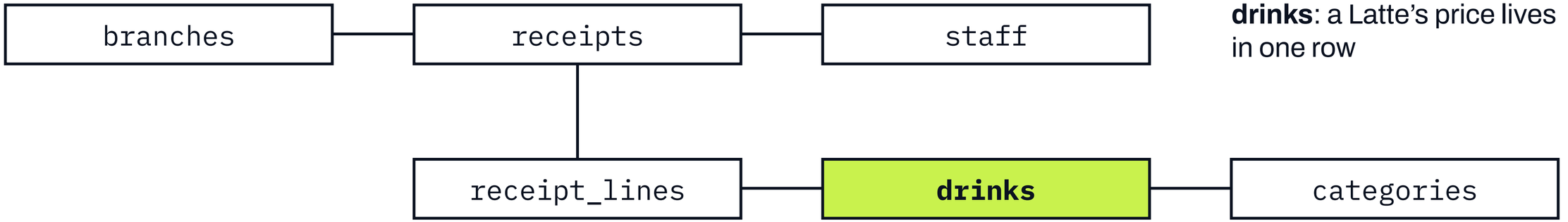
reads all 9,180 lines of October, groups them, adds up

The same rows serve both jobs. A shape that is quick to write one sale into is slow and awkward to ask questions of.

DEFINITION · OLTP · THE CAFÉ'S TILL · MADE-UP

OLTP: the system that records each sale as it happens

OLTP (online transaction processing) records each sale as it happens. It splits the data into many small tables, so each piece of information is stored once.



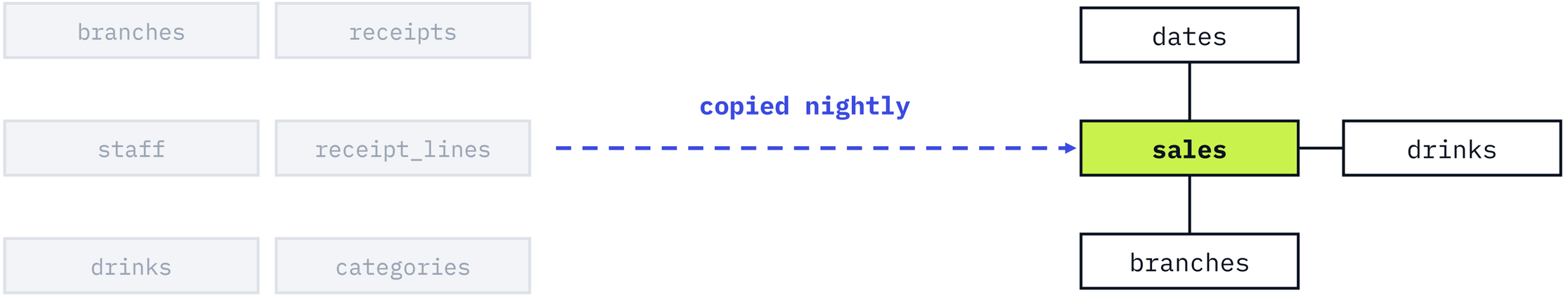
drinks: a Latte's price lives in one row

The same idea as D06: Lina's new city was one change in one row of customers.

Built for writing: each sale touches a few rows, and nothing is stored twice.

OLAP and the data warehouse: where questions are answered

OLAP (online analytical processing) answers questions by reading many rows and adding them up. A **data warehouse** is a copy of the data, shaped for those questions.



the till: shaped for writing

the warehouse: shaped for reading

The till stays fast, and every question reads the copy.

INSTRUCTOR DO

Step 1.2

Facts, dimensions and the star schema

Every business question is a number plus “by” words. The numbers become facts; the “by” words become dimensions.

MADE-UP QUESTIONS FROM THREE BUSINESSES

A business question is a number plus the words after “by”

café	**Revenue**	by	**month**	by	**drink category**	
clinic	**Patient visits**	by	**department**	by	**week**	
airline	**Seats sold**	by	**route**	by	**month**	

the number becomes a fact

the “by” words become dimensions

Read every request this way before you build anything: what is added up, and how it is sliced.

DEFINITION · FACT TABLE · THE CAFÉ'S sales TABLE · MADE-UP

A fact table records events: one row for each receipt line

A fact table records events. Each row is one receipt line (a drink and its quantity): keys that point at the dimension tables, and the numbers you add up.

date_key	drink_key	branch_key	qty	amount
2026-10-01	D1	B1	2	5.00
2026-10-01	D2	B1	1	1.50
2026-10-01	D4	B1	3	6.00
2026-10-01	D2	B1	2	3.00
2026-10-01	D1	B1	1	2.50

keys: which day, which drink, which branch

numbers: what you add up

grows: 9,180 rows in October, about 296 a day

Microsoft: "Fact tables store observations or events." Here, each receipt line is one event.

DEFINITION · DIMENSION TABLE · THE CAFÉ’S drinks AND branches · MADE-UP

A dimension table describes things: one row for each drink

A dimension table describes the things in the “by” words. Each row is one thing: a unique key, and the columns you group and filter by.

drinks

drink_key	name	category	size
D1	Latte	**Coffee**	Large
D2	Espresso	**Coffee**	Small
D3	Mint tea	**Tea**	Medium
D4	Lemon mint	**Cold drinks**	Large

branches

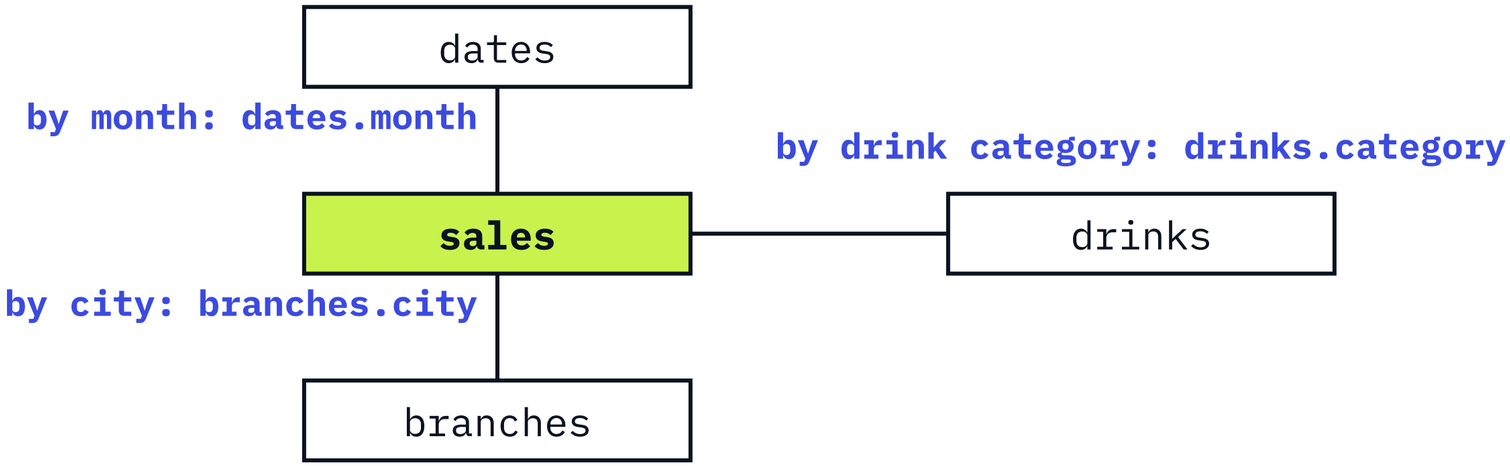
branch_key	city
B1	Amman
B2	Irbid
B3	Aqaba

category: “by drink category” groups the sales by this column

Four drinks and three branches: small tables that change slowly, while sales grows every day.

The café’s sales drawn as a star schema

A star schema puts the fact table in the middle and a dimension table around it for each “by” word, each joined to it by one key.



Sources: Kimball and Ross, The Data Warehouse Toolkit (2013); Microsoft, “Understand star schema and the importance for Power BI”.

Separate tables store each drink’s description once

ONE FLAT TABLE · 9,180 ROWS

date	drink	category	city	amount
10-01	Latte	**Coffee**	Amman	5.00
10-01	Espresso	**Coffee**	Amman	1.50
10-01	Lemon mint	Cold drinks	Amman	6.00
10-01	Espresso	**Coffee**	Amman	3.00

A STAR · THE drinks TABLE

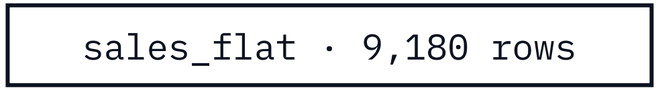
drink_key	name	category
D1	Latte	**Coffee**
D2	Espresso	**Coffee**
D3	Mint tea	Tea
D4	Lemon mint	Cold drinks

One flat table	rename Coffee: 5,372 rows change (every Latte and Espresso of October)	✗
A star	rename Coffee: 2 rows in drinks, and every report sees it	✓

Descriptions live in the dimension tables. The fact table keeps only keys and numbers, so it stays narrow as it grows.

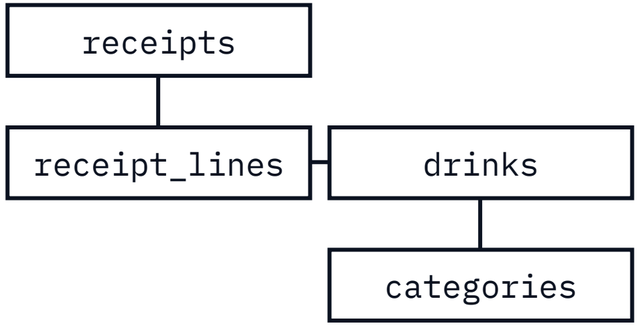
Three shapes of model for the same café sales

ONE FLAT TABLE

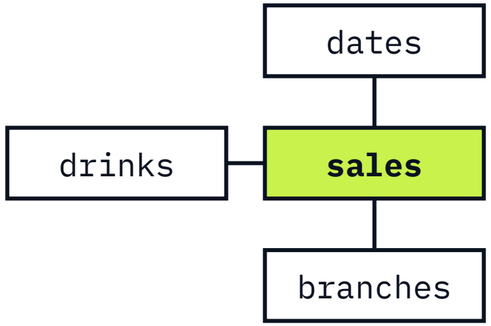


every row repeats the drink, category and city

THE TILL'S TABLES



A STAR

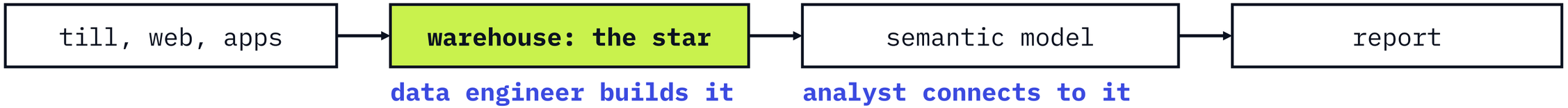


One flat table	no relationships; rename Coffee: 5,372 rows; fine for one small file	×
The till's tables	each piece of information once; revenue by category: lines → drinks → categories	×
A star	one step from sales to each dimension; rename Coffee: 2 rows	✓

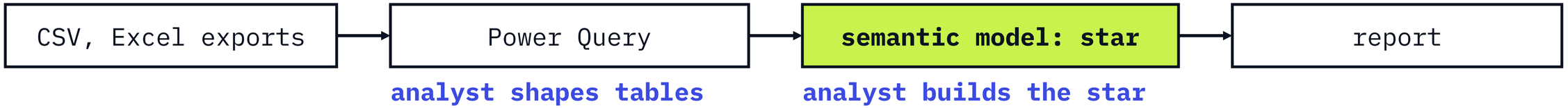
Power BI works with all three. Microsoft recommends the star for reports.

Who builds the star: a data engineer or the data analyst

WHEN THE COMPANY HAS A DATA WAREHOUSE



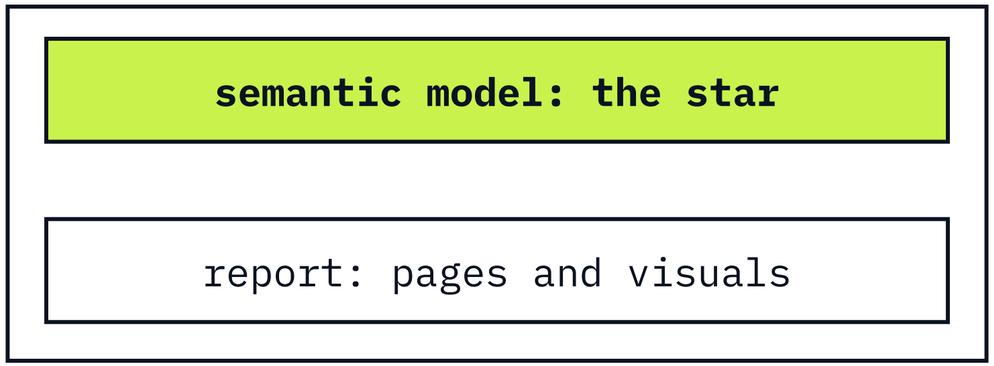
WHEN THE DATA ARRIVES AS FILES



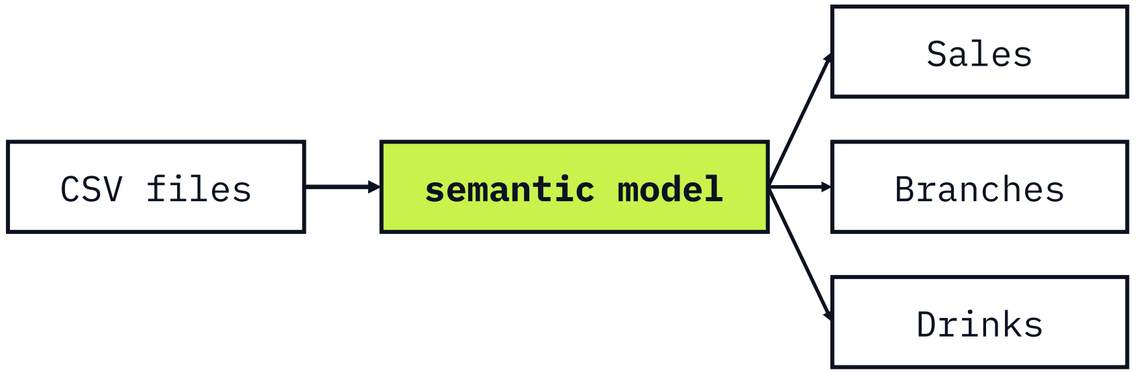
Data engineers usually build the star. Your job is to recognise facts and dimensions and connect them in Power BI.

One semantic model can serve many reports

POWER BI DESKTOP · ONE .pbix FILE



ON THE WEB · SEPARATE ITEMS · LINEAGE VIEW



three reports

Desktop	Publish: the file becomes a semantic model and a report on the web	✓
On the web	change the model once, and every report built on it sees the change	✓

Microsoft, “Separate reports from models in Power BI Desktop”.

QUICK CHECK

Which words are the fact and which are the dimensions?

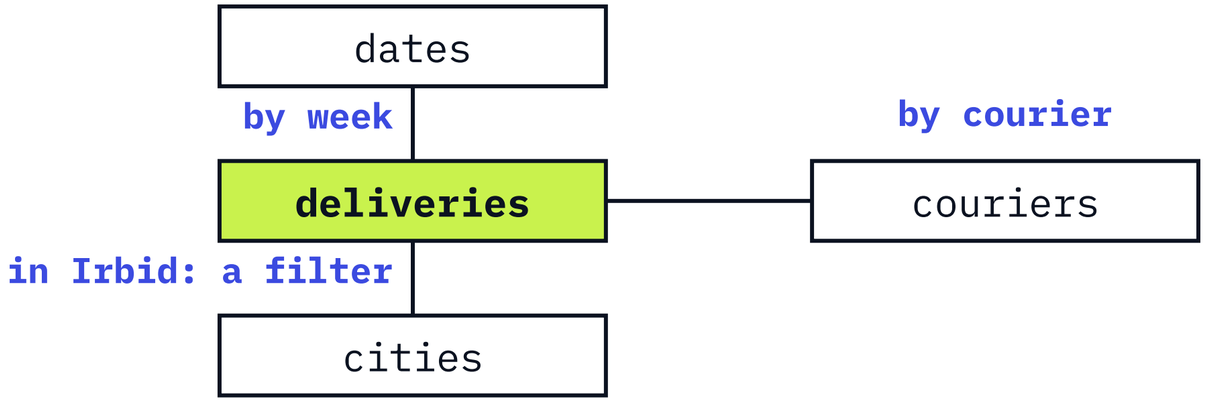
A delivery company asks: “What is the average delivery time by courier by week in Irbid?” Which part becomes the fact, which parts become dimensions, and what does “in Irbid” do?

THE ANSWER

Delivery time is the fact, and courier and week are dimensions

average delivery time by **courier** by **week** **in Irbid**

average says how to summarise delivery_minutes, the number in deliveries; it is not a new fact



“In Irbid” is a filter on a column of a dimension table, cities. It narrows the rows; it does not add a dimension to the result.

THE RULE TO KEEP

The numbers in a business question go in the fact table, and each “by” word is a column of a dimension table.

The star puts each dimension one step from the facts.

EVERYONE DO

Step 1.3

Warm-up quiz on D17’s Append, folder, Reference and Merge

Five questions, live in Markdown Mash: Append, the folder, Reference, the match count, and one on LEFT JOIN.

INSTRUCTOR DO

Step 1.4

From D17’s queries to today’s model

D17 merged the store list into the sales. Today the tables stay apart in the semantic model: the tables and the lines between them.

YOU KNOW THESE FROM SQL · TODAY IN THE MODEL

SQL you already know and its name in a Power BI model

SQL you wrote	The model today	Where
one row is one ___	the grain of a table	activity 01
a LEFT JOIN to a lookup table	a relationship, one to many	activity 03
NULL when the key finds nothing	a (Blank) row	activity 03
GROUP BY a lookup column	a visual by a dimension’s column	activity 03
COUNT(DISTINCT customer_id)	Count (Distinct)	activity 03

The anchor: completed revenue 316,023.50 (D16): the shops 205,338.00, online 110,685.50.

PART 2

Nakheel’s star: one fact table and its dimension tables

Order 100003 taken apart, the joins you wrote in SQL, and the grain of each table.



THE MODEL TODAY · PART 2
tables and grain

lime: this part · **white**: built
grey: later · dashed: inactive
arrow: the filter’s way

INSTRUCTOR DO

Step 2.1

The facts, dimensions and grain of Nakheel’s tables

Order 100003 taken apart, D06’s joins, D08’s views, then the grain and a number copied onto the wrong rows.

labs/D18/01-Evr-Load-Tables-Build-Sales-Lines/README.md

ONE ORDER, DRAWN · ORDER 100003, THE SECOND ROW OF D17'S APPEND PAGE

Order 100003 has one header row and three lines

orders · one row

order_id	customer_id	order_date	channel	order_total
100003	C-1357	2024-09-03	store	**47.00**



order_items · its three lines

item_id	product_id	qty	price	qty × price
500005	P-014	1	24.00	**24.00**
500006	P-060	1	18.00	**18.00**
500007	P-026	1	5.00	**5.00**

24.00 + 18.00 + 5.00
= 47.00, the header's order_total

One sale at two levels of detail: the order and its lines. Adding the order's total on every line counts it once per line.

DEFINITION · ONE ROW IS ONE ___

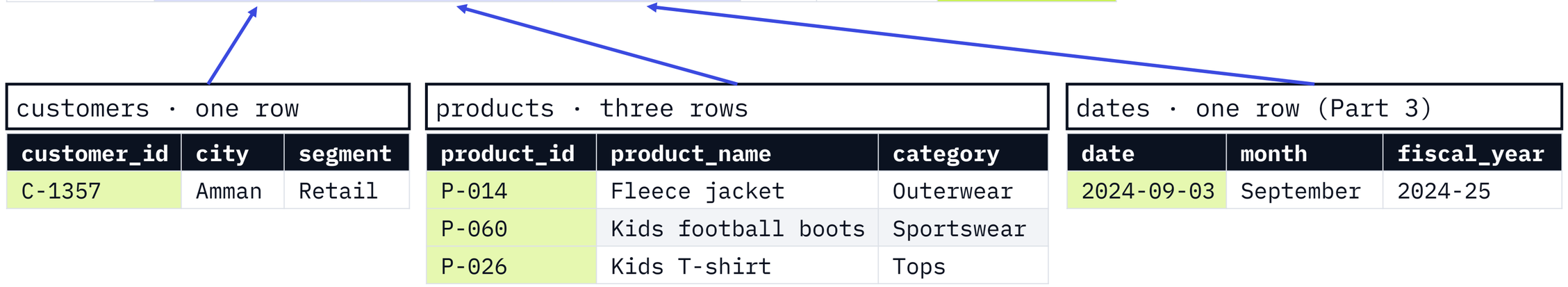
The grain of a table is what one row is. Say it as a sentence: one row is one order line. Every table in a model has exactly one.

Order 100003 as three fact rows and their dimension rows

sales_lines · three fact rows, one per line (activity 01)

order_id	customer_id	product_id	order_date	qty	price	line_value
100003	C-1357	P-014	2024-09-03	1	24.00	**24.00**
100003	C-1357	P-060	2024-09-03	1	18.00	**18.00**
100003	C-1357	P-026	2024-09-03	1	5.00	**5.00**

24.00 + 18.00 + 5.00 = 47.00
the order's total, now in three rows



customers · one row

customer_id	city	segment
C-1357	Amman	Retail

products · three rows

product_id	product_name	category
P-014	Fleece jacket	Outerwear
P-060	Kids football boots	Sportswear
P-026	Kids T-shirt	Tops

dates · one row (Part 3)

date	month	fiscal_year
2024-09-03	September	2024-25

Grouped by segment, all three lines fall in Retail; by category, in Outerwear, Sportswear and Tops; by month, in September 2024.

orders and order_items are fact tables

A fact table holds events, one row per sale or order line: keys that point at other tables, and numbers you add up.

customers · 500

customer_id	segment
C-1001	Retail
C-1002	Wholesale

one customer

products · 60

product_id	category
P-001	Shoes
P-014	Outerwear

one product

orders · 3,000

order_id	customer_id
100003	C-1357
100004	C-1335

one order

order_items · 7,516

order_id	product_id
100003	P-014
100003	P-060

one line of an order

things: next page

events: a new row with every order

Primary key Foreign key

orders and order_items both hold events: one order at two levels of detail, the order and its lines.

customers and products are dimension tables

A dimension table describes the things the events are about, one row per thing: a unique key, and the columns you filter and group by.

customers · 500

customer_id	segment
C-1001	Retail
C-1002	Wholesale

one customer

products · 60

product_id	category
P-001	Shoes
P-014	Outerwear

one product

orders · 3,000

order_id	customer_id
100003	C-1357
100004	C-1335

one order

order_items · 7,516

order_id	product_id
100003	P-014
100003	P-060

one line of an order

things: one row each; the key appears once

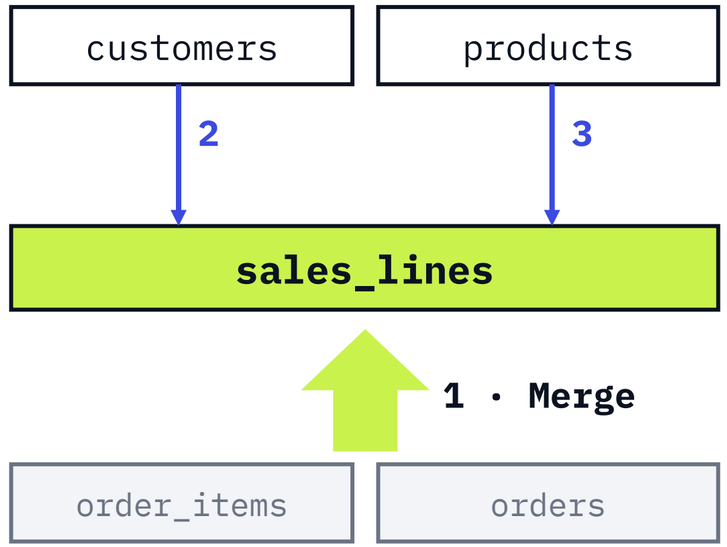
events: previous page

Primary key Foreign key

An ID does not make a dimension: orders has order_id, and it grows with every sale.

D06’s three joins become one Merge and two relationships

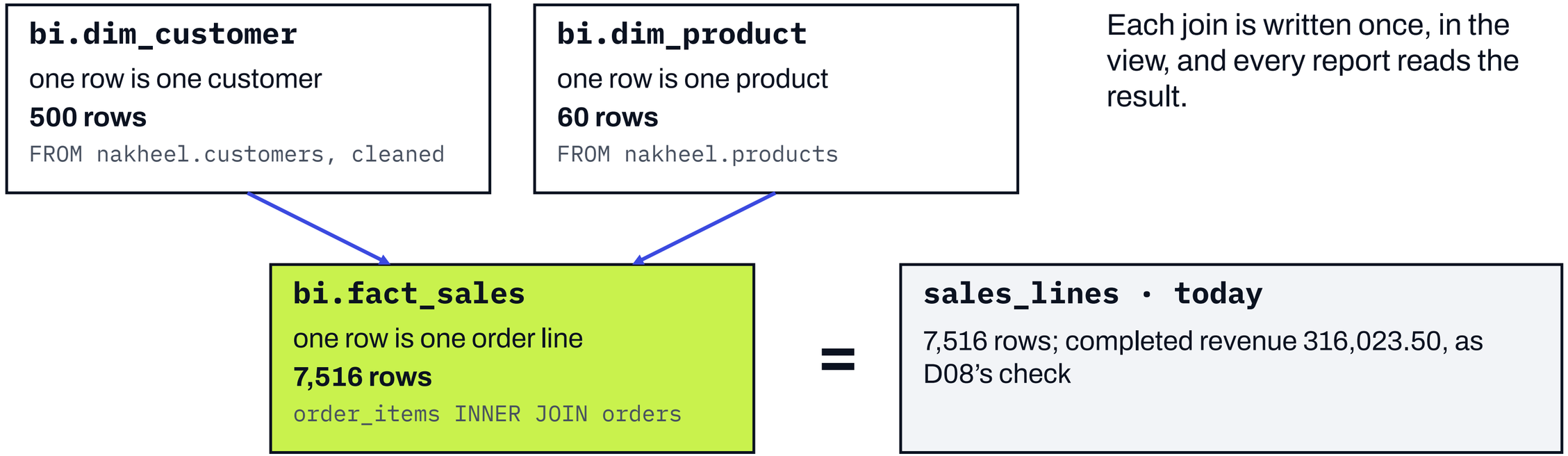
```
   SELECT c.segment, p.category,
          SUM(i.qty * i.price) AS revenue
   FROM nakheel.order_items AS i
1  INNER JOIN nakheel.orders AS o
     ON i.order_id = o.order_id
2  LEFT JOIN nakheel.customers AS c
     ON o.customer_id = c.customer_id
3  INNER JOIN nakheel.products AS p
     ON i.product_id = p.product_id
   WHERE o.status = 'completed'
   GROUP BY c.segment, p.category;
```



lime: one table at one grain (Merge)
indigo: a dimension to the fact (relationship)

In SQL every question repeats the joins. In the model, join 1 runs once in Power Query, and joins 2 and 3 run whenever a visual asks.

D08’s bi views are a star written in SQL



Each join is written once, in the view, and every report reads the result.

You wrote this star in SQL on D08. Today Power BI holds the same tables and the relationships between them.

THE PLANNED DEAD END · order_total MERGED ONTO THE LINES · COMPLETED ORDERS

order_total on every line counts the revenue 3.1 times over

Order 100003: 47.00 on each of its three lines adds up to **141.00**

Sum of order_total on the lines



988,971.50

Sum of line_value (qty × price)

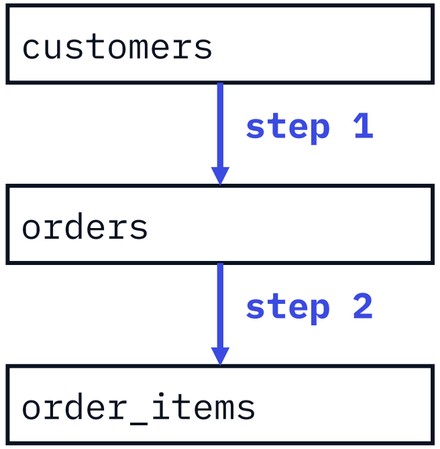
316,023.50

order_total, shipping_fee	describe an order (shipping_fee: its delivery charge): they stay off the line table	×
qty, price, line_value	describe a line: they add up to 316,023.50, D16's answer	✓

3.1×, more than the 2.5 lines an average order has (7,516 ÷ 3,000): big orders have more lines. Shipping: 1,742.50 on the lines against 1,032.50.

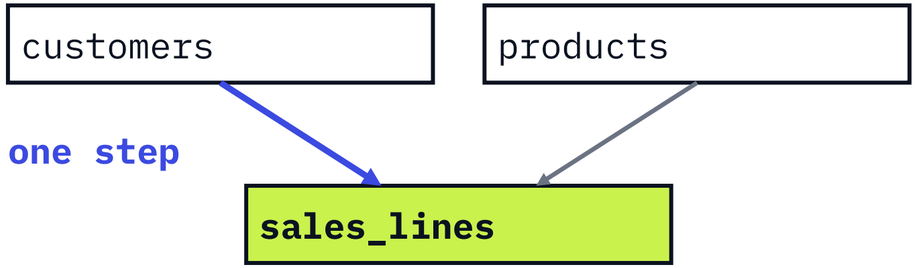
Two shapes for Nakheel’s orders: a snowflake chain or a star

SNOWFLAKE · tables in a chain, not around one fact



Indigo: a filter on segment travels two steps to reach a line; with products, four tables to load and relate.

STAR · orders merged into its lines



Indigo: the same filter in one step, as from every dimension. sales_lines: one row is one line, with the order’s customer and dates on it.

Merging here is safe because orders and order_items describe the same event; D17’s merge copied a dimension’s columns onto the sales.

EVERYONE DO

Step 2.2

Activity 01: build sales_lines and name the grain of each table

Load D16’s four CSV files, merge each order onto its lines, and say what one row is in every table.

labs/D18/01-Evr-Load-Tables-Build-Sales-Lines/README.md

WHAT COMES BACK · MERGE QUERIES AS NEW: order_items LEFT OUTER orders, ON order_id

sales_lines after the Merge: 7,516 rows, one per order line

order_id	product_id	qty	price	customer_id	order_date	status	line_value
100003	P-014	1	24.00	C-1357	2024-09-03	completed	**24.00**
100003	P-060	1	18.00	C-1357	2024-09-03	completed	**18.00**
100003	P-026	1	5.00	C-1357	2024-09-03	completed	**5.00**

From order_items: item_id, order_id, product_id, qty, price. **From orders:** customer_id, order_date, order_ts, delivered_date, channel, status. **Added:** line_value = qty × price (lime). Every line finds its order, so Left Outer and D08's INNER JOIN give the same rows.

Card: Sum of line_value	316,023.50 with a filter sales_lines[status] = completed: D16's answer	✓
Had order_total come across	988,971.50: order_total describes an order, not a line	×

ACTIVITY 01, STEP 5 · THE ANSWERS

The grain of each table in today’s model

Table	One row is one …	Rows	Kind
sales_lines	one line of an order	7,516	**fact**
customers	one customer	500	dimension
products	one product	60	dimension
dates	one day (Part 3)	1,096	dimension
orders	one order: inside sales_lines now	3,000	load off
order_items	one line: inside sales_lines now	7,516	load off

shipping_fee does not belong on sales_lines (1,742.50 on the lines against 1,032.50); price does, it is the line’s own price; customer_id does, as a key.

THE RULE TO KEEP

Keep order_total on orders, because it describes an order, not a line.

qty × price describes a line, so line_value is the number sales_lines adds up.

PART 3

A date table: one calendar for every date column in the model

Two dates on every sale and one on every customer: one calendar that the dates relate to.



THE MODEL TODAY · PART 3
a date table

lime: this part · **white**: built
grey: later · dashed: inactive
arrow: the filter's way

INSTRUCTOR DO

Step 3.1

Why the model needs its own date table, dates

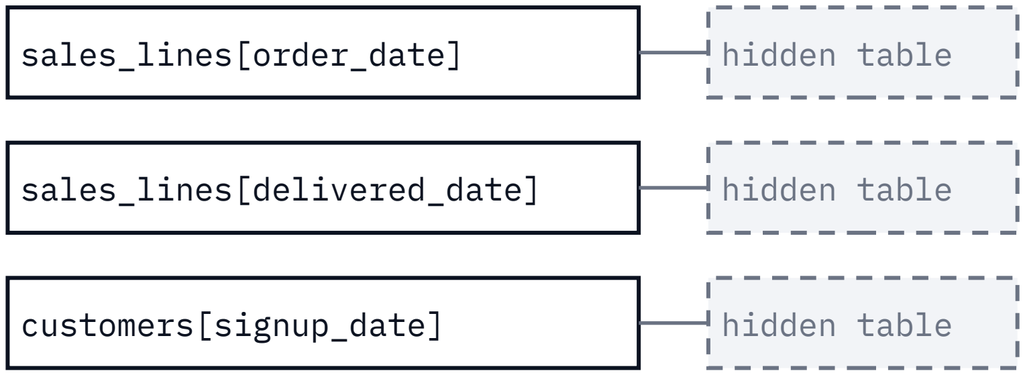
Hidden calendars before, one table after; month names in order; mark it as the date table.

labs/D18/02-Evr-Date-Table/README.md

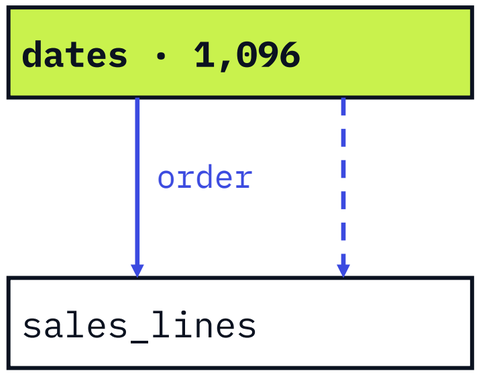
BEFORE · AUTO DATE/TIME (ON BY DEFAULT) → AFTER · ONE TABLE, dates

One dates table instead of a hidden calendar per column

BEFORE · a hidden calendar per date column



AFTER · one dates table, related to each



dashed: the delivered date, Part 5. signup_date is not related today, and gets no hidden table once Auto date/time is off.

A slicer (an on-page filter) on April 2026: which sales does it filter?

Hidden calendars	each column has its own April: a slicer on one leaves the others alone	✗
One dates table	one April slicer filters every visual of sales, by order date	✓

DEFINITION · ONE ROW IS ONE DAY

A date table has one row per day, with no day missing and none twice, and the columns reports group by: year, month, fiscal year. Every date column relates to it.

dates · FIVE OF ITS 1,096 ROWS · 2024-01-01 TO 2026-12-31

Five rows of dates and the columns reports group by

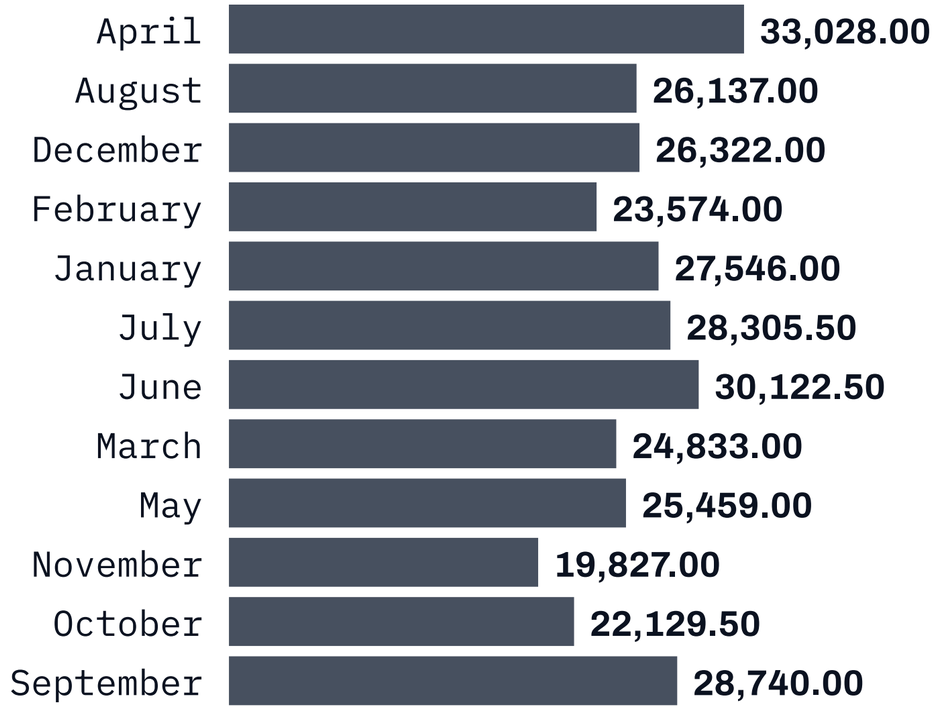
date	year	month_no	month	year_month	fiscal_year
2024-01-01	2024	1	January	2024-01	2023-24
2024-01-02	2024	1	January	2024-01	2023-24
2025-09-01	2025	9	September	2025-09	**2025-26**
2026-04-15	2026	4	April	2026-04	**2025-26**
2026-12-31	2026	12	December	2026-12	2026-27

Lime: fiscal_year runs September to August, so 1 September 2025 starts 2025-26. The data runs 2024-09-01 to 2026-08-31; the table covers whole calendar years, as Microsoft's guidance asks.

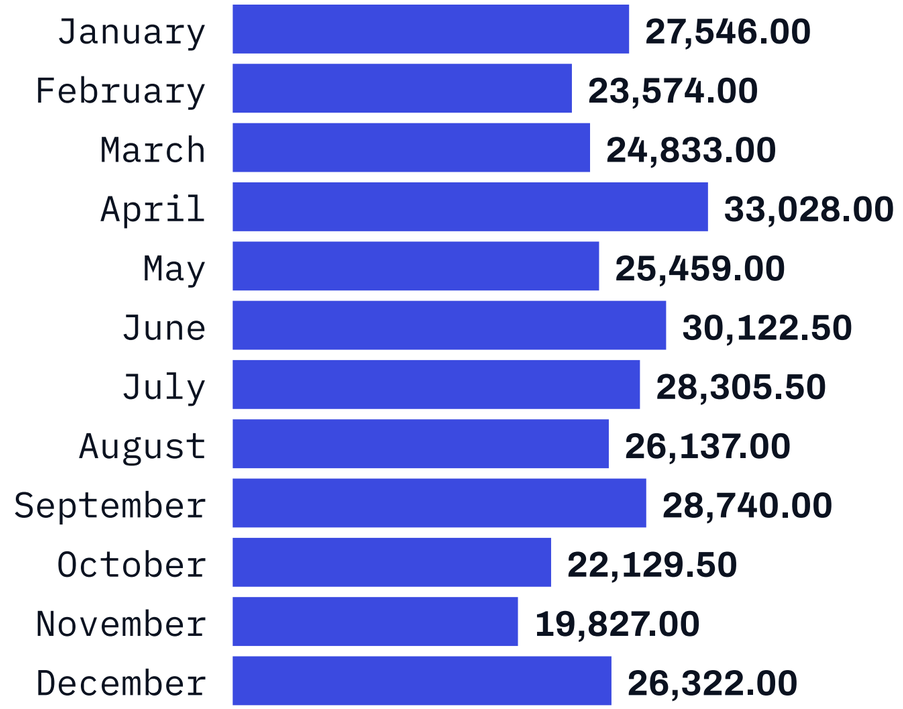
COMPLETED REVENUE BY month, ALL YEARS · BEFORE AND AFTER SORT BY COLUMN → month_no

Sort month by month_no so that January comes first

BEFORE · alphabetical: April and August first



AFTER · sorted by month_no



Sort by column orders month by month_no: one number per name. Here the two years are added together by month name.

Only dates[date] passes the four checks of Mark as date table

Check	dates[date]	sales_lines[order_date]	
		order_date	
Unique values	**1,096 rows, 1,096 dates**	**7,516 rows, 714 dates**	×
No null values	none	none	✓
Contiguous dates	**none missing**	**16 days missing**	×
Same time on all	type Date	type Date	✓

order_ts would fail the last check too: 2024-09-01 19:44:59 carries a time.

Mark dates	it becomes the model's calendar; D20's time functions read it	✓
Mark sales_lines[order_date]	refused: dates repeat and 16 days are missing	×

EVERYONE DO

Step 3.2

Activity 02: build, sort and mark the dates table

One calendar for every date in the model, its months in order, marked as the date table.

labs/D18/02-Evr-Date-Table/README.md

BEFORE → TO TABLE, ADD COLUMN → DATE → AFTER · A BLANK QUERY CALLED dates

From a list of 1,096 dates to the dates table

the formula bar of a Blank Query

```
= List.Dates(#date(2024, 1, 1), 1096, #duration(1, 0, 0, 0))
```

BEFORE · a list

List
2024-01-01
2024-01-02
2024-01-03
…



AFTER · a table, typed Date, five columns added

date	year	month_no	month	year_month	fiscal_year
2024-01-01	2024	1	January	2024-01	2023-24
2024-01-02	2024	1	January	2024-01	2023-24
2025-09-01	2025	9	September	2025-09	2025-26
…	…	…	…	…	…

Indigo: five columns worked out from date, three with Add Column → Date and two with Custom Column (year_month, fiscal_year).

PART 4

Relationships: how a filter on a dimension reaches the fact

Relate each dimension to the fact table, and see what each line lets a report answer.



THE MODEL TODAY · PART 4
relationships

lime: this part · **white**: built
grey: later · dashed: inactive
arrow: the filter’s way

INSTRUCTOR DO

Step 4.1

A relationship, its cardinality and its direction

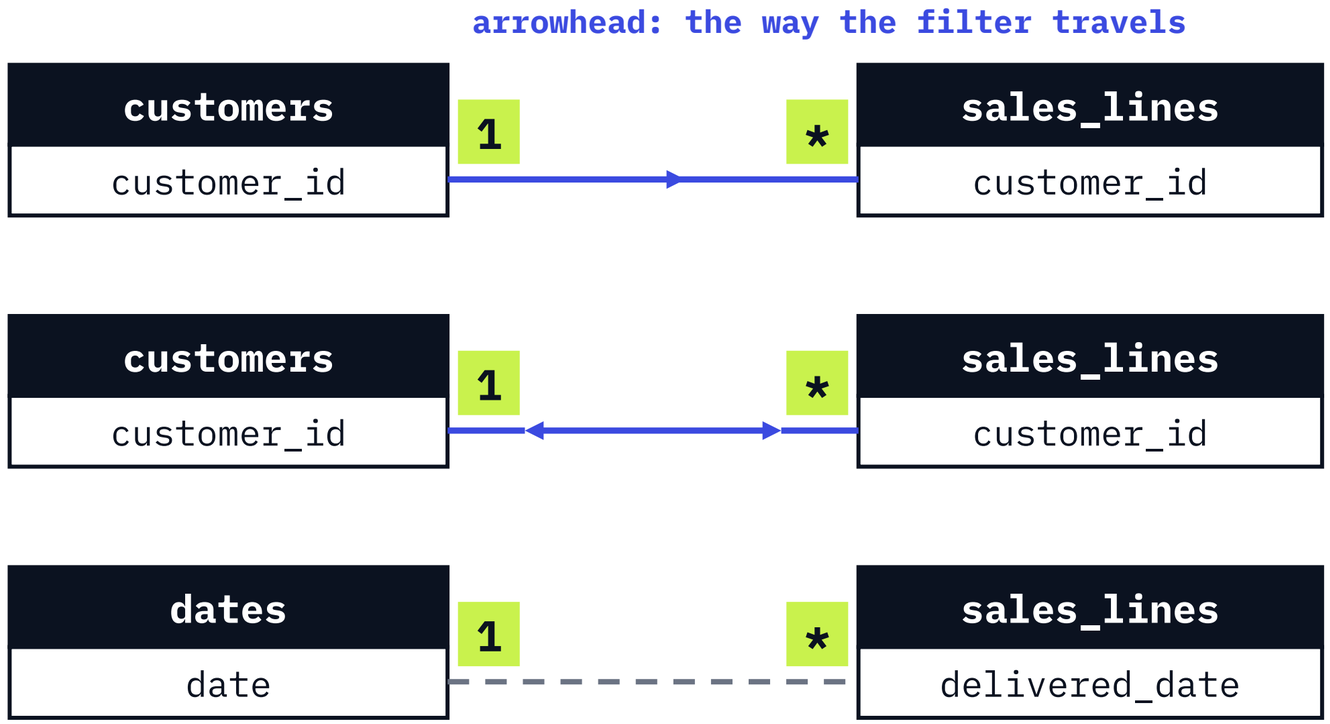
The marks on a line, cardinality on one order, D06’s match board, the blank row, and the star.

labs/D18/03-Evr-Relationships-And-Questions/README.md

DEFINITION · RELATIONSHIP

A relationship is a line between a key in a dimension table and the same key in a fact table. A visual that groups by the dimension’s column sends its filter along the line.

How Model view draws a relationship and its marks



Single: a segment picked in customers filters those customers' lines. Every line today is Single.

Both: the lines can filter customers back. It changes other visuals too (optional lab 06).

Dashed: inactive. No visual uses it until a measure, a named calculation, asks for it (Part 5).

1: the key is unique there. customers: 500 rows, 500 different codes.

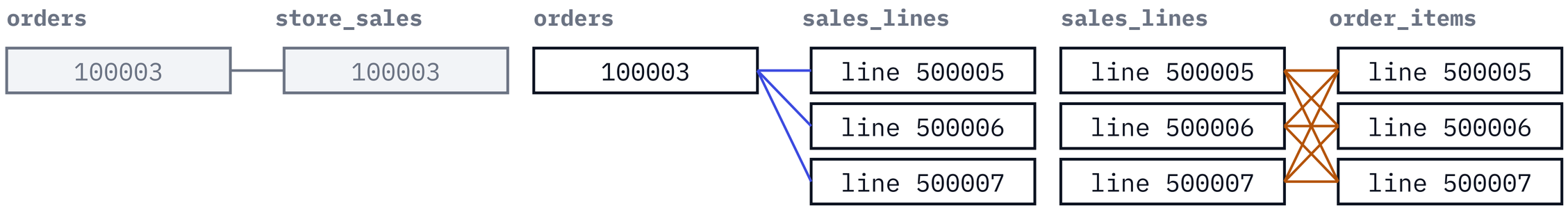
*: it repeats. sales_lines: 7,516 lines, 462 different codes.

Cardinality says how often a key value appears on each side of a relationship. One to many: once in the dimension, any number of times in the fact.

CARDINALITY · THE SAME KEY, order_id = 100003, IN THREE PAIRS OF TABLES

Three kinds of cardinality on order 100003

One to one (1:1)	**One to many (1:*)**	Many to many (*:*)
once on each side (D17's store_sales, not in today's model)	once here, three times there: the kind of every relationship in today's star	three on each side: neither table describes the other

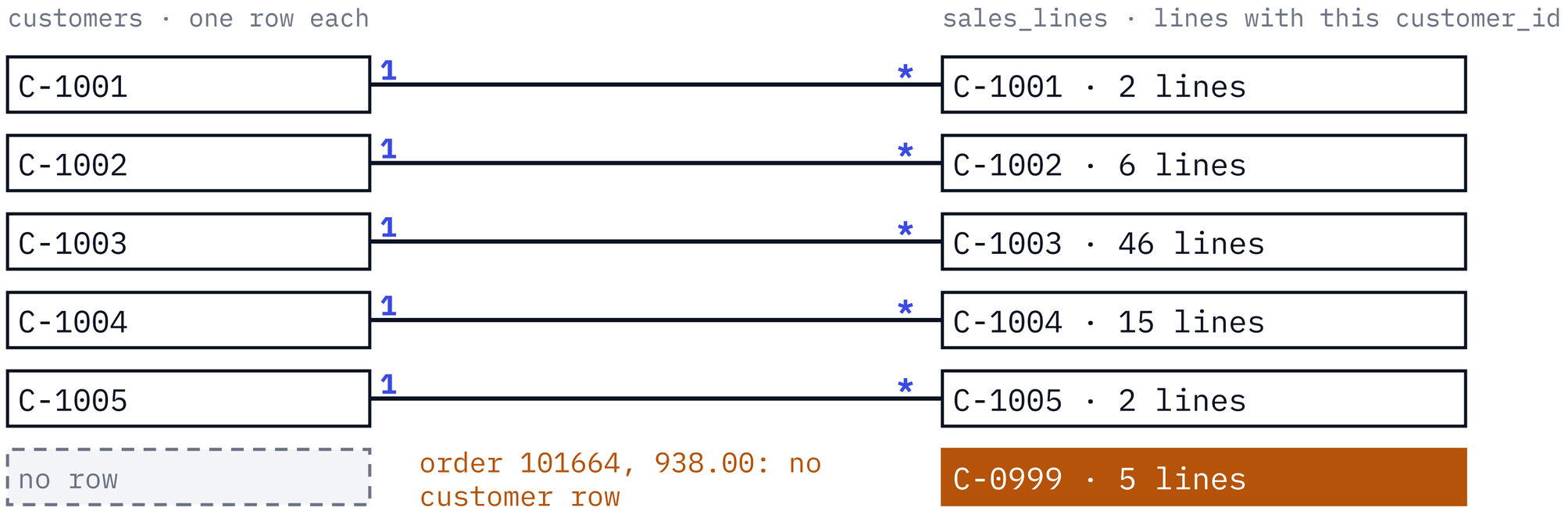


each box is one row with order_id 100003; each connector joins two rows on that key

a dimension's key	unique in the dimension, so every star line is one to many	✓
a key twice in a dimension	D17's doubled store code: the line becomes many to many	×

D06’S MATCH BOARD · customers[customer_id] → sales_lines[customer_id]

customers to sales_lines is one to many



Microsoft: the one side “contains unique values”; the many side “can contain duplicate values”. C-0999 (amber) is on the lines but not in customers.

WHAT COMES BACK · COMPLETED REVENUE BY customers[segment]

The order with no customer appears as (Blank)

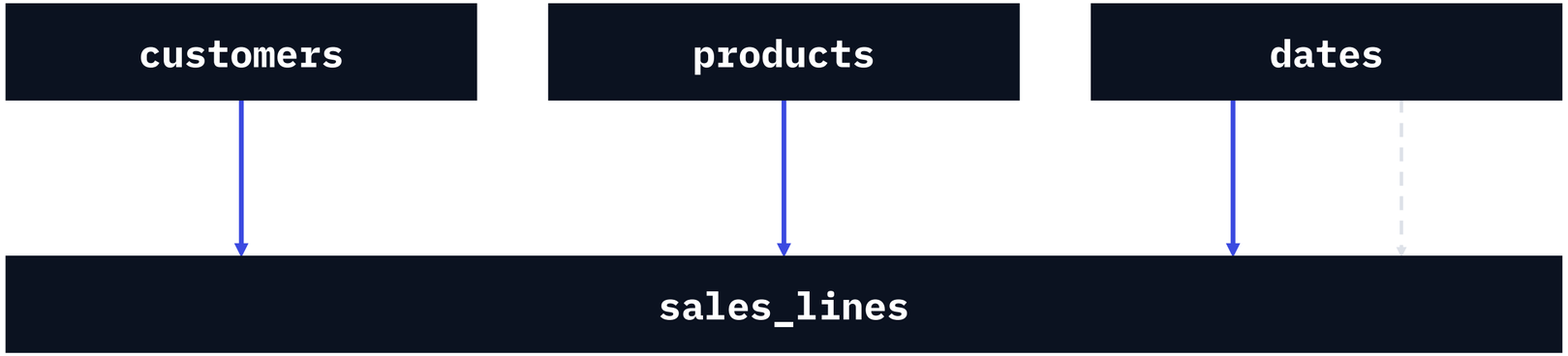


Microsoft: “a blank virtual row is added to the ‘one’ side table” for a value with no match.

together 316,023.50, D16’s answer: no line is lost

(Blank) = order 101664, customer C-0999, five lines: the same order D06 and D07 found. D17’s stall was a (Blank) governorate the same way.

The star: each relationship adds one way to group revenue



`customers → sales_lines`	by segment	**160,881.50 Retail · 154,204.00 Wholesale**	✓
`products → sales_lines`	by category	**Outerwear · 76,637.00 first**	✓
`dates → sales_lines`	by month	**2026-04 best · 20,237.00**	✓

Each chart adds up to 316,023.50. The faint dashed line comes after the break.

EVERYONE DO

Step 4.2

Activity 03: relate the tables and answer the manager’s questions

Three lines into sales_lines, then revenue by segment, by category and by month.

labs/D18/03-Evr-Relationships-And-Questions/README.md

INSTRUCTOR DO

Step 4.3

What a Single filter direction cannot answer

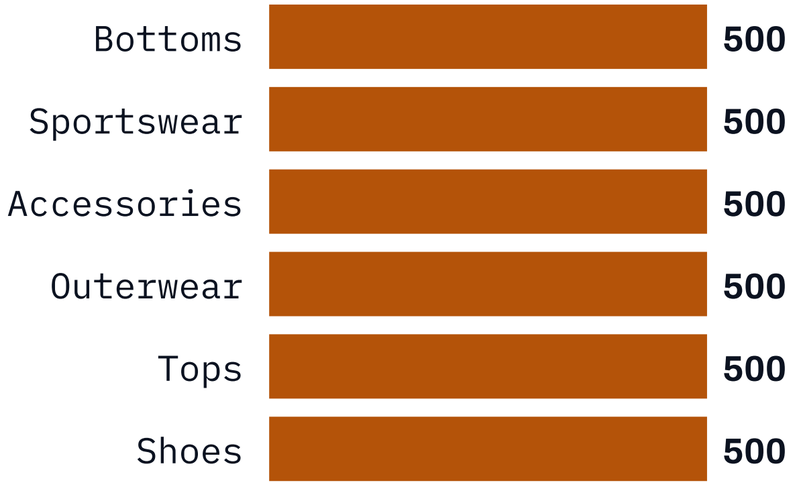
“How many customers bought Shoes?” Two charts, one wrong on every bar, and two ways to fix it.

labs/D18/03-Evr-Relationships-And-Questions/README.md

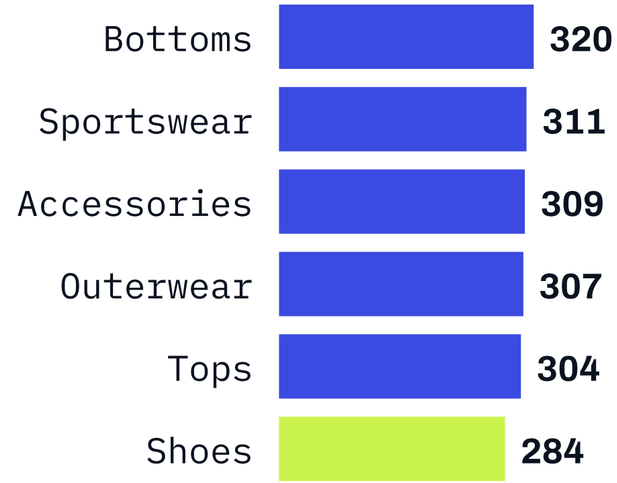
Cross filter direction is the way the filter travels. Single: from the one side to the many side. Both: either way.

Customers by category reads 500 on every bar

Count of customers[customer_id]



Count (Distinct) of sales_lines[customer_id]





indigo: the Shoes filter's path. It reaches sales_lines and cannot go on to customers

Left: all 500 customers on every bar. Right: count where the filter arrives, 284 bought Shoes.

ONE QUESTION, THREE ROUTES · “HOW MANY CUSTOMERS BOUGHT SHOES?”

Three ways to count the customers who bought Shoes

`Count customers[customer_id]`	500 on every bar: customers is never filtered	✗
`Count (Distinct) sales_lines[customer_id]`	284, with every relationship Single	✓
`Cross filter direction: Both`	283 customer rows, but every visual of customers now follows product filters (lab 06)	✗

The same number on every row means the filter cannot reach the table you are counting. In SQL: COUNT(DISTINCT customer_id) after the join.

284 counts every buyer code on the lines, C-0999 included; 283 counts rows of customers, which has no C-0999. Microsoft: “minimize the use of bi-directional relationships”.

products[product_name] AND SUM OF line_value · BEFORE AND AFTER “SHOW ITEMS WITH NO DATA”

Show items with no data brings back unsold products

BEFORE

56

products

a visual shows only the rows the fact table has values for

AFTER

60

products

four come back with no value (amber): D07’s products that never sold

product_id	product_name	category	line_value
P-009	Hiking boot	Shoes	**(blank)**
P-025	Graphic T-shirt	Tops	**(blank)**
P-046	Sunglasses	Accessories	**(blank)**
P-058	Headband	Sportswear	**(blank)**

Without the status filter, 39 customers come back the same way: D07’s customers who never ordered.

QUICK CHECK

Which way does the filter run?

products 1 → * sales_lines, Single. A chart groups revenue by category. Does the filter travel from products to sales_lines, or from sales_lines to products? And what would a count of products by segment show?

THE ANSWER

The filter runs from products, the one side, to sales_lines



the arrowhead points at the fact table

Revenue by category	the filter on a category reaches its lines: right	✓
Count of products by segment	60 on every bar: segment cannot reach products	×

The fix is the same as for Shoes: count product_id in sales_lines with Count (Distinct).

THE RULE TO KEEP

A filter runs from the one side of a relationship to the many side.

From the thing to its events. To count the things, count their key in the fact table.

EVERYONE DO

Step 4.4

Activity 03, continued: the manager’s questions

Five questions, each answered with one visual and one relationship. Learners who fall behind finish them from the README.

labs/D18/03-Evr-Relationships-And-Questions/README.md

ACTIVITY 03

The manager’s questions for activity 03

INSTRUCTIONS

1. Retail or wholesale: who brings more? 2. Online only: the top category? 3. The best month? 4. How many different customers bought Sportswear? 5. Which products never sold? For each, name the relationship the answer uses.

Full steps and Check yourself: labs/D18/03-Evr-Relationships-And-Questions/README.md

REVIEW

Time's up.
Let's review.

Activity 03: the manager’s questions

`Solved: labs/D18/03-Evr-Relationships-And-Questions/Solved/solution.md`

The answers to activity 03 and the relationship each one uses

1 · customers	Retail 160,881.50, Wholesale 154,204.00, (Blank) 938.00	✓
2 · products	online: Outerwear 25,315.50, just ahead of Shoes 25,120.50	✓
3 · dates	the best month: 2026-04, 20,237.00	✓
4 · products	Sportswear: 311 different customers (Count (Distinct))	✓
5 · products	Hiking boot, Graphic T-shirt, Sunglasses, Headband	✓

channel in question 2 is a column of sales_lines itself: a filter, no relationship.

Break

Stretch, then two dates

After the break: the delivered date, a tidy model, and Lab 04.

PART 5

The order date and the delivered date on one fact table

Sold in April and delivered in April are different questions. One date table answers both.



THE MODEL TODAY · PART 5
two dates

lime: this part · **white**: built
grey: later · dashed: inactive
arrow: the filter’s way

INSTRUCTOR DO

Step 5.1

One date table in two roles: the order date and the delivered date

Delivery times, April sold against April delivered, a solid line and a dashed one, and the measure.

labs/D18/04-Lab-Two-Dates-Tidy-Model/README.md

DAYS FROM ORDER TO DELIVERY · DELIVERED ORDERS, BY CHANNEL

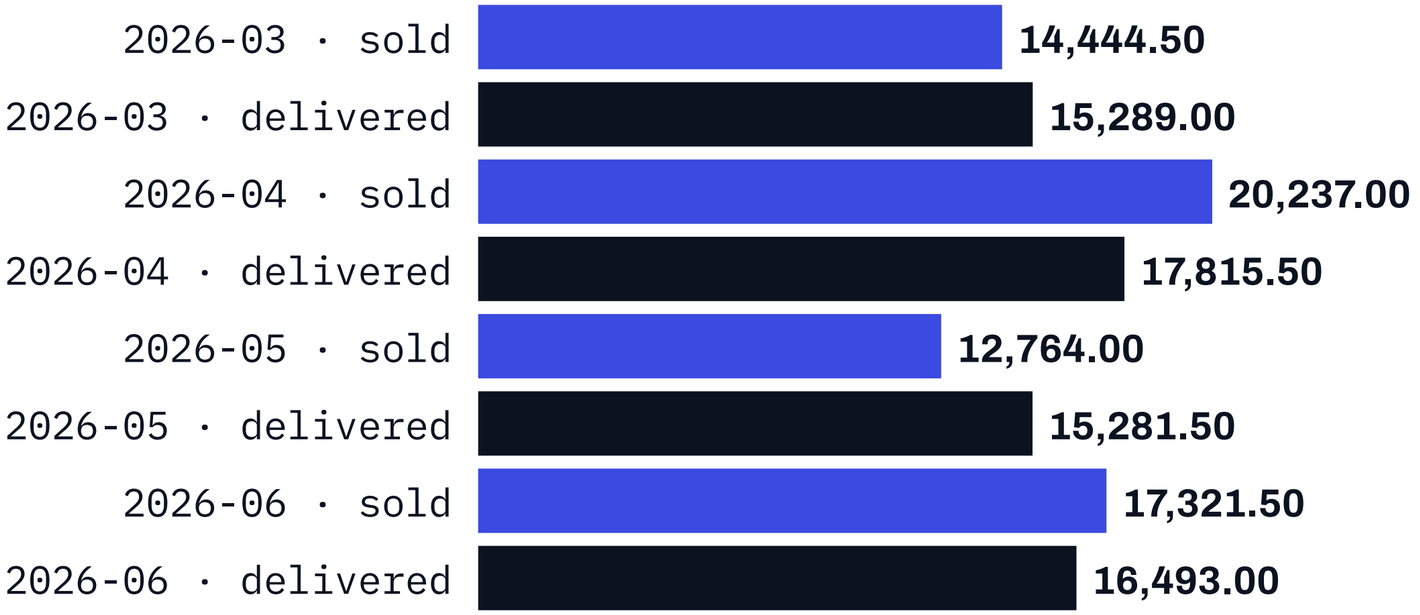
Online orders take one to four days to deliver



Cancelled (212) and pending (17) orders have no delivered_date: only completed orders were delivered.

COMPLETED REVENUE, MARCH TO JUNE 2026 · BY ORDER MONTH AND BY DELIVERED MONTH

Sold and delivered revenue, March to June 2026



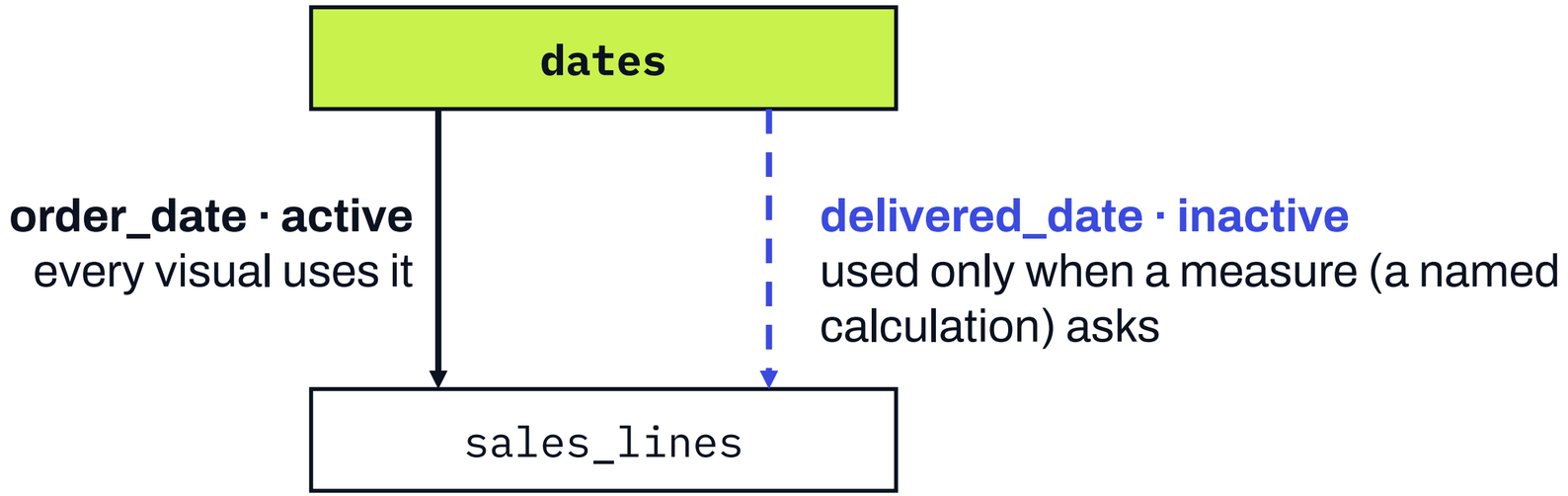
April’s orders: 17,524.50 delivered in April, 2,712.50 in May

Two questions, two numbers: which one the manager means is a choice you make in the model.

A role-playing dimension is one dimension table used in more than one role. dates plays the order date and the delivered date.

MODEL VIEW · dates RELATED TO sales_lines TWICE

dates has two lines to sales_lines, one inactive



dates[year_month] on an axis means the order month. Power BI keeps one active path between two tables, so the second line is inactive: one month cannot mean two dates at once.

Two ways to report revenue by delivered date

`dates[year_month]` on the axis	20,237.00: the order month, not the delivery month	✗
A measure with `USERELATIONSHIP`	17,815.50: the dashed line, inside one measure (D19 preview)	✓
A second table, `delivery_dates`	17,815.50: an active line of its own; costs a second table	✓

Today: the measure. Microsoft says the inactive relationship is fine when no visual needs both dates at once; when one does, copy the date table.

A PREPARED MEASURE · Resources/delivered_revenue.dax

A prepared measure that uses the inactive line

A measure is a named calculation (D19). This one adds up line_value along the delivered-date line.

```
Delivered revenue =
CALCULATE(
    SUM(sales_lines[line_value]),
    USERELATIONSHIP(sales_lines[delivered_date], dates[date])
)
```

April 2026 → 17,815.50

PART 6

A tidy model: hide the keys and remove unused columns

What a report author should see in the Data pane, and two columns nobody uses. Then Lab 04.



THE MODEL TODAY · PART 6
a tidy model

lime: this part · **white**: built
grey: later · dashed: inactive
arrow: the filter's way

INSTRUCTOR DO

Step 6.1

Hide the keys and remove the columns no report uses

What a report author should see in sales_lines, and two columns nobody uses.

labs/D18/04-Lab-Two-Dates-Tidy-Model/README.md

sales_lines fields before and after hiding keys and dates

BEFORE · 12 fields and 1 measure

item_id	order_date
order_id	order_ts
product_id	channel
qty	status
price	delivered_date
customer_id	line_value

AFTER · 6 fields and 1 measure

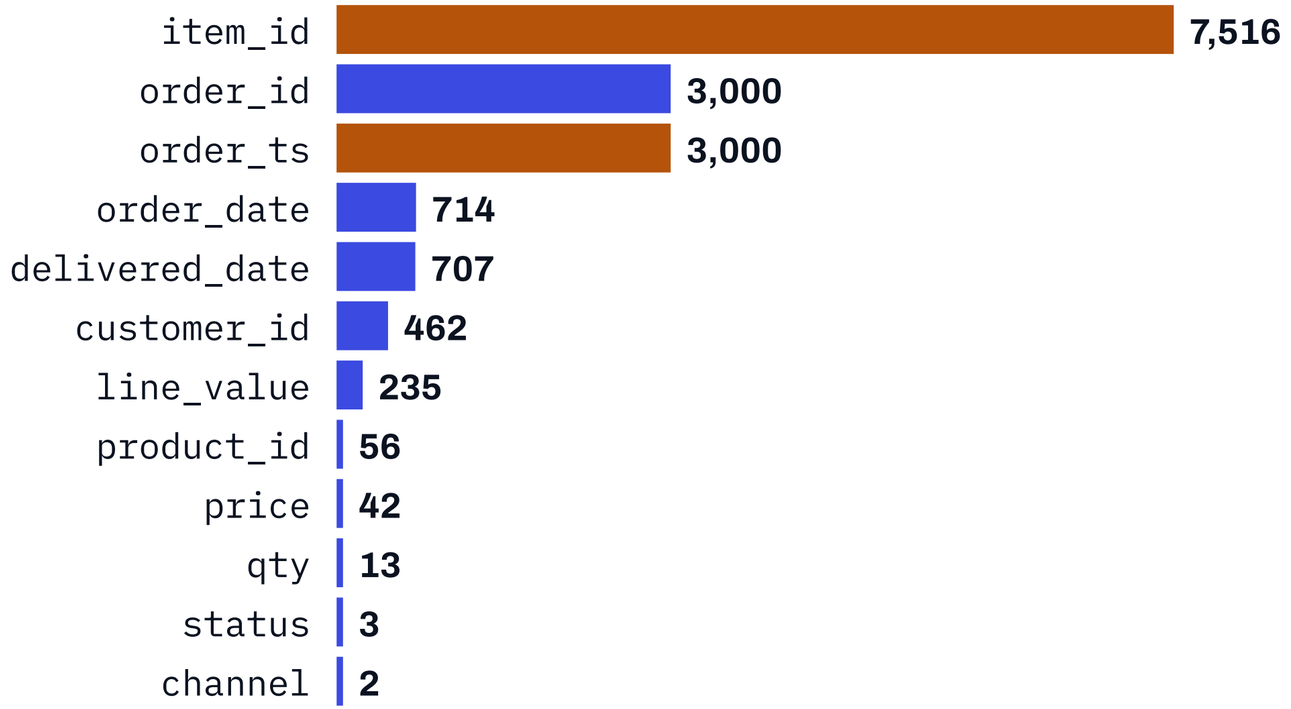
order_id	**channel**
product_id · hidden	**status**
qty	delivered_date · hidden
price	**line_value**
customer_id · hidden	
order_date · hidden	

Delivered revenue the measure from Part 5: in sales_lines before and after, never hidden

amber: removed in Power Query (next page) · **grey**: hidden, still used by the relationships

DISTINCT VALUES PER COLUMN OF sales_lines · 7,516 ROWS

Remove item_id and order_ts, which nothing uses



The model stores each column's distinct values once. **item_id** (every row different) and **order_ts** (date and time) are used by no visual and no relationship: remove them in Power Query.

Removed in Power Query, never loaded; hidden in the model, still used. order_id stays: D19 counts orders with it.

EVERYONE DO

Step 6.2

Lab 04: two dates and a tidy model

Relate the delivered date, paste Delivered revenue, then remove two columns and hide the keys. Finish on your own if we run out of time.

labs/D18/04-Lab-Two-Dates-Tidy-Model/README.md

REVIEW

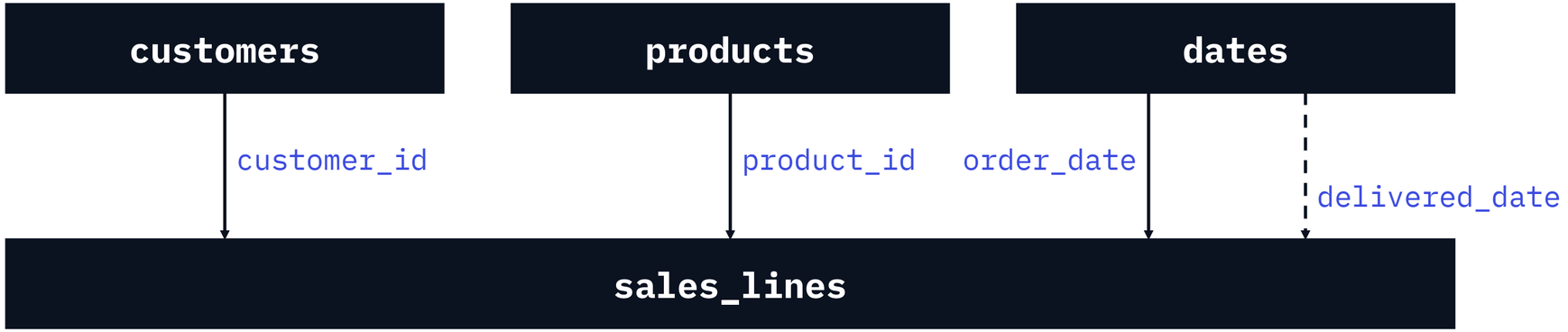
Time's up. Let's review.

Lab 04: two dates and a tidy model

`Solved: labs/D18/04-Lab-Two-Dates-Tidy-Model/Solved/solution.md`

THE MODEL YOU BUILT TODAY · ONE FACT TABLE, THREE DIMENSION TABLES

The finished model: a star with one-to-many Single relationships



solid: active · **dashed:** inactive, used by Delivered revenue · **arrowhead:** the filter’s direction

Each label names the key the relationship joins on. Every chart by segment, category or month adds up to 316,023.50 (D16).

THE RULE TO KEEP

Hide the keys and the columns that report authors should not use.

Remove what no visual and no relationship uses, in Power Query, before it loads.

STUDY LIST · IN CLASS AND IN THE LABS

Today’s Power BI features and where to practise each

In class	What it does	Activity
Merge queries as new	order header onto its lines: sales_lines	01
List.Dates, To Table	a date table, one row per day	02
Sort by column	month names in calendar order	02
Mark as date table	dates becomes the model’s calendar	02
Model view: drag a key	a relationship, one to many	03
Count (Distinct)	count where the filter arrives	03
Show items with no data	rows with no sales come back	03
USERELATIONSHIP	a measure uses the inactive line	Lab 04
Is hidden	a field disappears from the Data pane	Lab 04

Lab	After class, self-guided, not submitted
04	**Two dates and a tidy model: finish it if class ran out of time**
05	**Optional: the shops in the same model (the appendix pages)**
06	**Optional, for PL-300: cross filter direction Both and its cost**
07	**Optional, for PL-300: targets by month, after lab 05**

CLOSE

What you can do now in Power BI

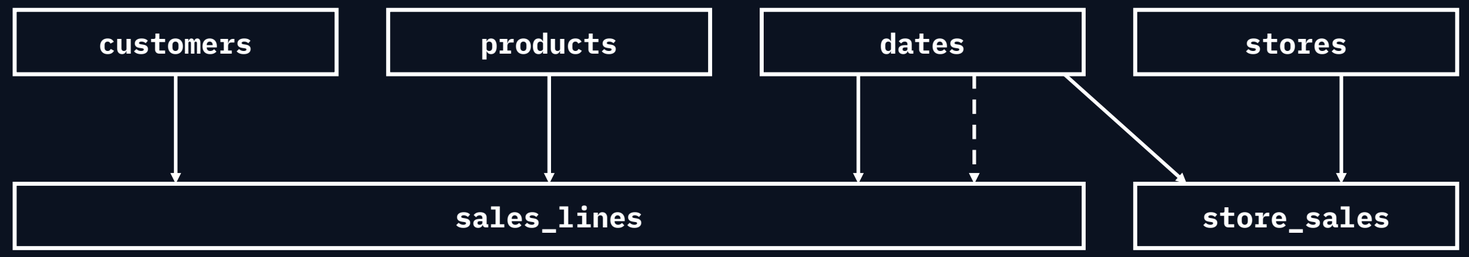
- Name the grain of each table
- Build and mark a date table
- Relate tables and read 1, * and the arrow
- Use a second date and hide keys

NEXT · D19 Measures: SUM, then CALCULATE.

LAB 05

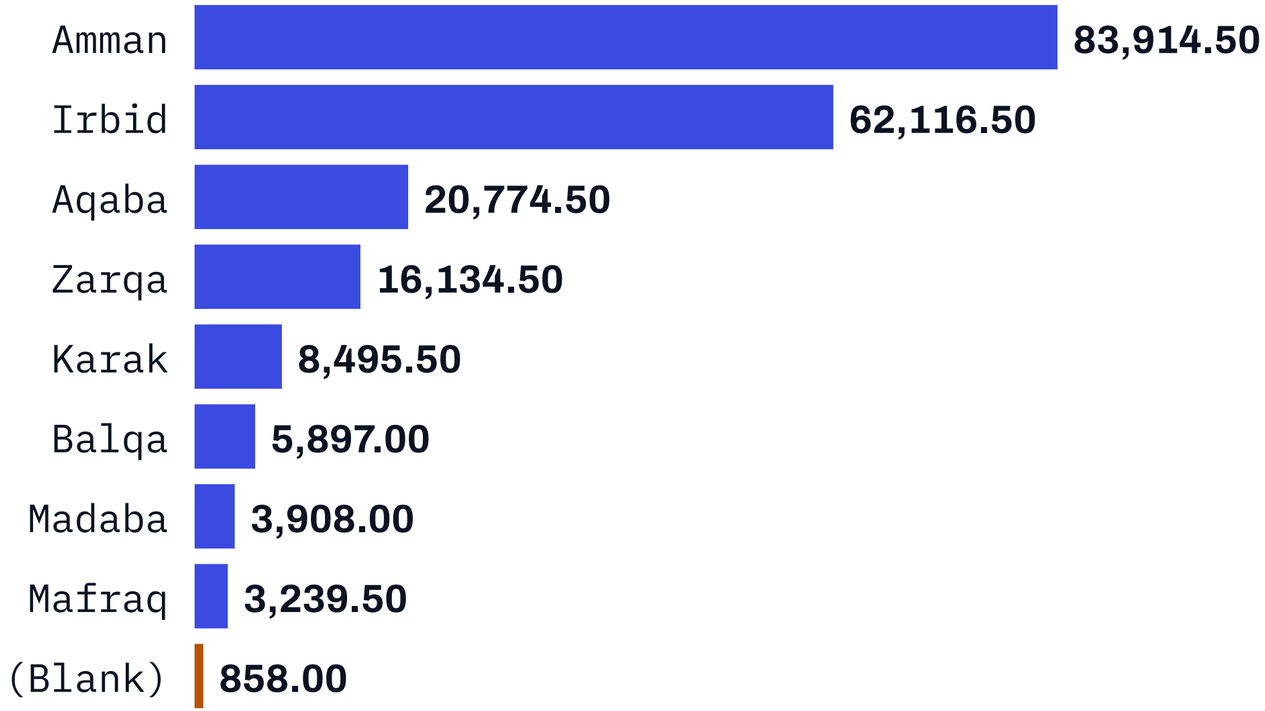
For the optional lab 05: the shops in the same model

D17’s store list and till sales join the star, and two fact tables need care.



LAB 05 · stores[governorate], SUM OF store_sales[order_total], COMPLETED

D17’s governorate bars come back through a relationship



(Blank) is NK-POP-01, the Ramadan stall: in no store list.

Jerash has a row in stores and no sales: Show items with no data adds it.

Power Query compared codes with their case on D17. Microsoft says the model engine “is case insensitive”; D17 fixed the key in stores, so this line matches either way.

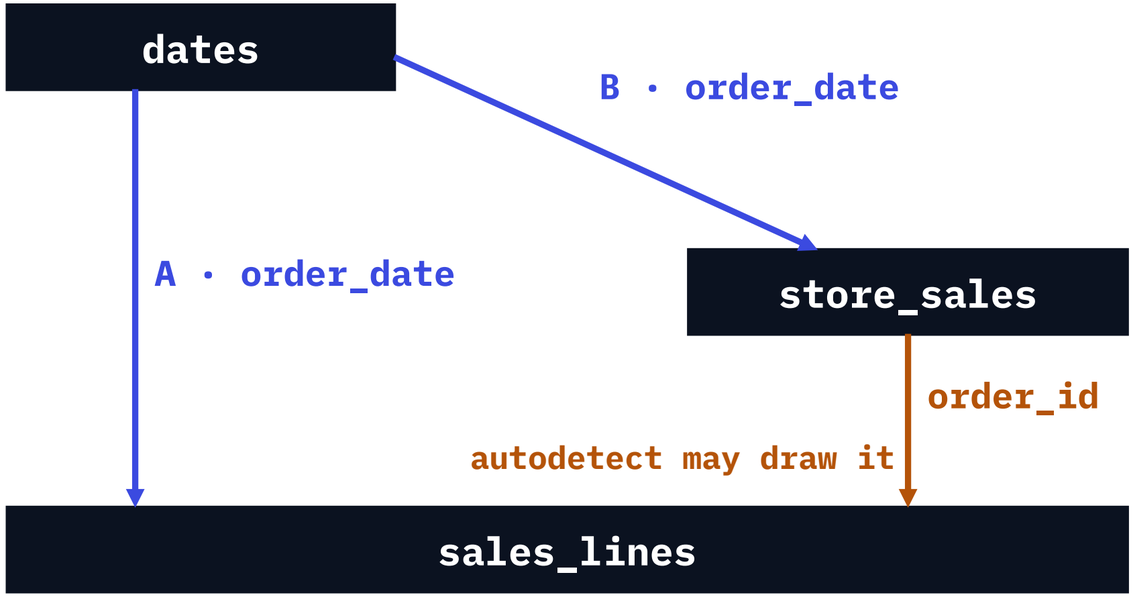
stores reaches store_sales but not sales_lines



Store revenue by month	store_sales and sales_lines (channel = store) agree: April 2026 13,585.50 in both	✓
Revenue by governorate	from store_sales: D17's bars. From sales_lines (store): 205,338.00 on every row	×
Store revenue by category	not in this model: stores does not reach sales_lines	×

Store orders are in both tables, at two grains: never add the two together. Shop questions: store_sales; product questions: sales_lines.

Two paths from dates to the same lines give two April totals



A 2026-04 slicer, completed lines:

A · direct

20,237.00 every line ordered in April

B · via store_sales

13,585.50 only lines of a store order

Online orders (6,651.50) have no store order, so path B never finds them.

Delete the order_id line	dates reaches each fact table directly: one answer	✓
Keep it	one path goes dashed; if it is dates–store_sales, months stop filtering the shops	×